

MD MUSHFIQUR RAHMAN

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Jackson, Mississippi 39202

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EDUCATION

University of Mississippi Medical Center 2024– Present
PhD, Department of Data Science Jackson, Mississippi
Third-year student (currently enrolled). Research focus: Multistate Modelling, Survival Analysis.

University of Mississippi Medical Center 2024 – 2026
MS, Biostatistics and Data Science Jackson, Mississippi

University of Dhaka 2018 – 2023
BS, Department of Statistics Dhaka, Bangladesh

EXPERIENCE

MIND Center, University of Mississippi Medical Center 2024 – Present
Student Biostatistician (Part-Time)

- Provide statistical support for neuroscience and aging research, with a focus on gait speed, cognitive status, and disease progression modeling.
- Apply functional regression, multistate models, and longitudinal methods to complex biomedical datasets.
- Collaborated on the project “Associations of Alzheimer’s Disease and Cerebrovascular Disease Biomarkers with Cognitive, Mobility, and Dual Impairment in the ARIC Study” (2025), presented by MSTAR student Meredith Boles.
- Currently working on the **MCSA (Mayo Clinic Study of Aging)** and **ARIC** studies, modeling transitions across cognitive states (normal cognition, MCI, dementia) using multistate survival models.
- Collaborate with clinical researchers on study design, analysis, and scientific reporting.

icddr,b October 2023 – June 2024
Data Management Assistant

- Organized and managed large datasets ensuring accurate and efficient data processing.
- Utilized STATA, SPSS, and R for storage, analysis, and visualization of clinical data.
- Assisted in several interim analyses of clinical trials to assess the efficacy of multiple new drugs.
- Analyzed Diarrheal Disease Surveillance System (DDSS) data at the Dhaka Hospital.
- Participated in a Phase III double-blinded randomized controlled trial using a probiotic to monitor weight gain for moderately and severely stunted children (2 months to 1 year of age).
- Built a Shiny app to showcase and analyze various aspects of the study.

Save the Children in Bangladesh May 2023 – July 2023
Research Assistant

- Performed analysis to measure maternal and newborn care indicators (ANC visits, essential newborn care, place and mode of delivery, postpartum family planning).
- Conducted multivariate analysis on factors contributing to ANC utilization, mode of delivery, and newborn care practices.
- Assessed intervention effectiveness using the LiST (The Lives Saved Tool).
- Managed survey data and created a data dictionary.

Statistical Research Consultancy Bangladesh (SRCBD)

November 2020 – July 2021

Trainee Data Analyst (Internship)

- Aided in course curriculum design for R programming.
- Performed data visualization using `ggplot2` and hypothesis testing in R.

RESEARCH & PUBLICATIONS

- [1] Hassan, Md. Mehedi and Mamun Billah, Md. Al and **Rahman, Md. Mushfiqur** and Zaman, Sadika and Hasan Shakil, Md. Mehadi and Angon, Jarif Huda. **Early Predictive Analytics in Healthcare for Diabetes Prediction Using Machine Learning Approach**. *2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, 2021, pp. 01–05. DOI: [10.1109/ICCCNT51525.2021.9579799](https://doi.org/10.1109/ICCCNT51525.2021.9579799).
- [2] Hassan, Md. Mehedi and Hassan, Md. Mahedi and Akter, Laboni and **Rahman, Md. Mushfiqur** and Zaman, Sadika and Khan, Md. Hasib and Jahan, Nusrat and Smrity, Raisun Nasa and Farhana, Jerin and Raihan, M. and Mollick, Swarnali. **Efficient Prediction of Water Quality Index (WQI) Using Machine Learning Algorithms**. *Human-Centric Intelligent Systems*, vol. 1, pp. 86–97, 2021. DOI: [10.2991/hcis.k.211203.001](https://doi.org/10.2991/hcis.k.211203.001).
- [3] Hassan, Md. Mehedi and Raihan, M. and Khan, Md. Hasanul Banna and Dhali, Tutul and **Rahman, Md. Mushfiqur** and Sneha, Zarmina Huda. **A Hybrid Machine Learning Approach to Predict the Risk of Having Stroke**. *ACM*, 2022, pp. 460–465. DOI: [10.1145/3542954.3543020](https://doi.org/10.1145/3542954.3543020).
- [4] Hassan, Md. Mehedi and Zaman, Sadika and **Rahman, Md. Mushfiqur** and Bairagi, Anupam Kumar and El-Shafai, Walid and Rathore, Rajkumar Singh and Gupta, Deepak. **Efficient Prediction of Coronary Artery Disease Using Machine Learning Algorithms with Feature Selection Techniques**. *Computers and Electrical Engineering*, vol. 115, 109130, 2024. DOI: [10.1016/j.compeleceng.2024.109130](https://doi.org/10.1016/j.compeleceng.2024.109130).
- [5] Md Fuad Al Fidah and Md Ridwan Islam and **Rahman, Md. Mushfiqur** and Mohammad Jobayer Chisti and ASG Faruque and Tahmeed Ahmed and Sharika Nuzhat. **Community-Level Expenses for Under-5 Diarrhoeal Cases Prior to Reporting at Dhaka Hospital: A Cross-Sectional Analysis in Bangladesh**. *Clinical Epidemiology and Global Health*, vol. 31, 101846, 2025. DOI: [10.1016/j.cegh.2024.101846](https://doi.org/10.1016/j.cegh.2024.101846).
- [6] Bashir, Syed Jayedul and Islam, Md Ridwan and Nuzhat, Sharika and Amin, Rukaeya and **Rahman, Md. Mushfiqur** et al. **Antibiotic Use Prior to Attending a Large Diarrheal Disease Hospital Among Preschool Children Suffering from Bloody or Non-Bloody Diarrhea: A Cross-Sectional Study Conducted in Bangladesh**. *PLOS ONE*, 19(11): e0314325, 2024. DOI: [10.1371/journal.pone.0314325](https://doi.org/10.1371/journal.pone.0314325).
- [7] Nuzhat, Sharika and Islam, Md Ridwan and Fidah, Md Fuad Al and Islam, Shoaib Bin and **Rahman, Md Mushfiqur** and Paul, Sneha and Nesa, Mahfuz-Un and Chowdhury, Devjani and

Nabi, Shayontan Fardin and Awn, Awni Mokhtar Sallam Ali and Khan, Liaquat Ali and Kawser, Chowdhury Ali and Chisti, Mohammad Jobayer and Ahmed, Tahmeed. **Maternal Knowledge, Attitude and Practice Regarding Commercial Oral Rehydration Salt Solution: Experience from a Diarrhoeal Disease Hospital in Bangladesh.** *BMJ Paediatrics Open*, 9(1): e003299, 2025. DOI: [10.1136/bmjpo-2024-003299](https://doi.org/10.1136/bmjpo-2024-003299).

[8] Nuzhat, Sharika and Rahman, Md. Mushfiqur et al. **Trends in Preventive Practices Against Diarrhoeal Disease Among Under-Five Children: Experience from the Largest Diarrhoeal Disease Hospital in Bangladesh.** *BMJ Paediatrics Open*, 9(1): e003259, 2025. [PMC11979600](https://doi.org/10.1136/bmjpo-2024-003259).

[9] Rahman, Amrina and Rahman, Md. Mushfiqur and Yasmin, Farhana. **Revolutionizing Healthcare through Federated Learning.** In: *A Secure and Trusted Approach for Federated Learning*, Chapter 1. Taylor & Francis / CRC Press, 2024. [Publisher Link](#).

PRESENTATIONS & CONFERENCES

- **Quantifying Neurocognitive Effects of Sleep Stage Architecture in OSA: A Variable Domain Functional Regression Approach.** *ASA Alabama–Mississippi Chapter.* 2026
2nd Place — Best Presentation Award.
- **Annual Meeting.** ASA Alabama–Mississippi Chapter. 2025
- **70th Annual Meeting.** Mississippi Academy of Science. 2025
- **69th Annual Meeting.** Mississippi Academy of Science. 2024

PROJECTS & REPORTS

- **Associations of Alzheimer’s Disease and Cerebrovascular Disease Biomarkers with Cognitive, Mobility, and Dual Impairment in the ARIC Study.** MIND Center, UMMC (collaborative project with MSTAR student Meredith Boles, presented 2025). 2025
- **Early Childhood Development: Variation within Age at Marriage and Birth of Children among Women in Bangladesh.** BS Final Year Project, University of Dhaka. 2022
- **Socio-Demographic Factors Affecting Maternal Health Care in Bangladesh.** 2020

DATA ANALYTICS SKILLS

Programming Languages	R, STATA, SPSS, SAS, MATLAB, Python, PowerBI
R Packages	ggplot2, dplyr, tidyr, caret, tidymodels, shiny, Quarto, survival, maptools
Software & Tools	Word, L ^A T _E X, Excel, PowerPoint

ACHIEVEMENTS

2nd Place — Best Presentation Award 2026
SOPH Research Day, University of Mississippi Medical Center.

Government Honours Scholarship 2024
Directorate of Secondary and Higher Education, People’s Republic of Bangladesh.

Best Paper Award

2022

Human-Centric Intelligent System, Springer.

Winner, Data Analysis Competition

2020

QMH Statistics Club, Department of Statistics, University of Dhaka.